

AI Governance Framework

Designing Strategic AI vs. Rules-Based Decision Logic for Enterprise SaaS

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~50% MANUAL OPS REDUCED	2 Features LLM TO RULES REDIRECTS	Zero HALLUCINATION RISK
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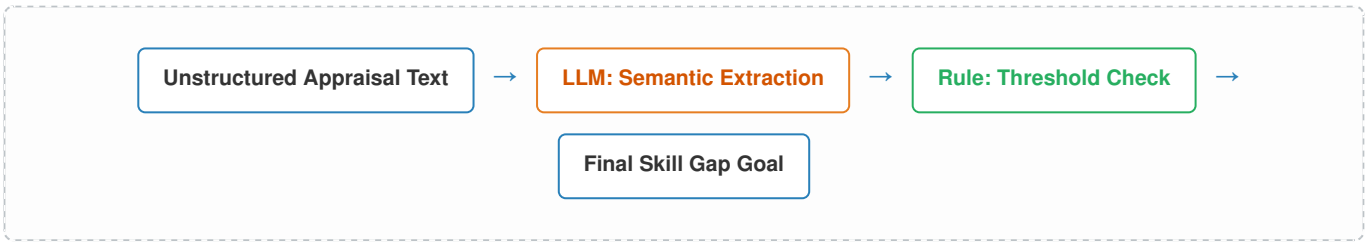
THE PROBLEM

As Akrivia HCM began integrating Generative AI, we faced several "AI-first" pitfalls that threatened delivery timelines and data integrity:

- **"AI for the Sake of AI":** Engineering teams defaulted to expensive, high-latency LLMs for logic easily solved with deterministic code.
- **Accuracy Risks:** Critical HR data (like skill gaps) requires objective outcomes, whereas LLMs are probabilistic and prone to hallucinations.
- **Operational Friction:** Without a review gate, AI insights reached employees directly, causing distrust if data was misaligned.

THE RESEARCH

Analyzing the **AI Skill-Gap workflow**, I identified that while LLMs excel at *semantic understanding* (reading unstructured appraisal text), they struggle with *deterministic thresholds* (verifying if a score meets a hard requirement).



THE DESIGN (THE FRAMEWORK)

- **Rule vs. LLM Playbook:** Established a decision tree: Use LLMs for subjective language tasks; use rules for objective data calculations.
- **Hybrid Workflow:** Architected a multi-stage flow where LLMs handle skill extraction, but rules-based logic governs the final output.
- **Human-in-the-Loop (HITL):** Implemented an HRBP Review Gate where administrators audit AI outputs before they become employee goals.
- **Evaluation Metrics:** Defined KPIs for prompt latency, token cost, and grounding accuracy.

BEFORE VS. AFTER

Dimension	Before (LLM-Only)	After (Governance Framework)
Feature Scoping	LLM as default for all features.	Redirected 2 features to rules-based.
Risk Profile	High risk of data hallucinations.	Hallucinations eliminated via rule gates.
Eng. Efficiency	Infinite "Prompt Tuning" cycles.	Significant engineering cycles saved.
User Ops	100% manual review needed.	~50% reduction in manual ops.

POST-LAUNCH CHALLENGE & FIX

Challenge: Processing large employee batches through LLMs caused synchronous timeouts and high DB pressure.

Fix: Introduced an **Async/Batch Architecture**. Requests are offloaded to background workers, and the UI updates via a notification once the "Review Gate" is ready for HR audit.

LESSONS CARRIED FORWARD

- **Governance Saves Innovation:** Prioritizing rules for simple tasks preserves budget for truly transformative AI use cases.
- **Deterministic > Probabilistic:** In B2B SaaS, accuracy is the primary feature. Hybrid models outperform pure-AI models in enterprise settings.
- **Review Gates Build Trust:** HITL design ensures HRBPs remain the final authority, increasing platform adoption.